

Reaction mechanism of the reconstituted oxoglutarate carrier from bovine heart mitochondria

Cesare INDIVERI¹, Thomas DIERKS², Reinhard KRÄMER² and Ferdinando PALMIERI¹

¹ Department of Pharmaco-Biology, Laboratory of Biochemistry and Molecular Biology, University of Bari and Consiglio Nazionale delle Ricerche Unit for the Study of Mitochondria and Bioenergetics, Italy

² Institut für Biotechnologie des Forschungszentrums, Jülich, Federal Republic of Germany

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The transport mechanism of the reconstituted oxoglutarate carrier, purified from bovine heart mitochondria, was studied kinetically. A complete set of half-saturation constants (K_m) was established for the two different substrates oxoglutarate and malate on both the external and the internal sides of the membrane. The internal affinities for oxoglutarate (K_m 0.17 mM) and malate (K_m 0.7 mM) were higher than the corresponding external affinities (K_m 0.3 mM and 1.4 mM, respectively). The exclusive presence of a single transport affinity for each substrate on one side of the membrane indicated a unidirectional insertion of the oxoglutarate carrier into the liposomal membrane. The K_m values and also the maximum exchange rates ($8–11 \mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$) for oxoglutarate and malate were independent of the nature of the counter substrate on the other side of the membrane. Under these defined conditions we analyzed the antiport mechanism in two-reactant initial velocity studies varying both the internal and external substrate concentrations. From the kinetic patterns obtained, a sequential type of mechanism became evident, implying that one internal and one external substrate molecule form a ternary complex with the carrier before transport occurs. A quantitative analysis of substrate interaction with the unloaded or single-substrate-occupied carrier revealed that rapid-equilibrium random conditions were fulfilled, characterized by a fast and independent binding of internal and external substrate. This kinetic mechanism agrees with previous results obtained in intact mitochondria. Considering also the data available for other mitochondrial carriers, a common kinetic mechanism (sequential type) for this carrier family is suggested.

The oxoglutarate carrier of mitochondria plays a central role in the transfer of reducing equivalents across the mitochondrial membrane. Together with the aspartate/glutamate carrier, it constitutes the membrane part of the malate/aspartate shuttle, the most important redox connection between the cytosol and the mitochondrial matrix. In first investigations in intact mitochondria from liver [1, 2] and heart [3], the carrier's substrate and inhibitor specificity, as well as some basic kinetic properties, were characterized. The transport system catalyzes an electroneutral 1:1 exchange of oxoglutarate and some other dicarboxylates, of which malate is bound with higher affinity. Detailed kinetic studies have been carried out in rat heart mitochondria by Sluse et al. (for reviews, see [4, 5]), leading to an exceptional complexity in two aspects. While saturation of the carrier with external malate [6], or with internal oxoglutarate [5], could be described by a normal Michaelis-Menten curve, this was not possible for internal malate due to negative cooperativity and due to allosteric influences of other metabolites [5, 7, 8]. In the case of external oxoglutarate a complex saturation curve was also observed [9], which was interpreted to reflect an oligomeric structure of the carrier protein [4–6].

In recent years, the oxoglutarate carrier has been purified from heart and liver mitochondria [10, 11]. After functional reconstitution into liposomes, it was confirmed that the basic transport properties of the isolated protein, i.e. substrate specificity and inhibitor sensitivity, agreed with the situation found in intact mitochondria [10–12]. Recently, the amino acid sequence of the oxoglutarate carrier has been determined by cDNA sequencing [13]. The primary structure was found to consist of a sequence repeated three times, as observed also for three other mitochondrial carrier proteins, the ADP/ATP carrier, the phosphate carrier and the uncoupling protein. The repeats of all four proteins clearly are interrelated [13].

Due to the methodological advantages of a reconstituted system, especially when using purified proteins, we now are able to elucidate the function of the oxoglutarate carrier under well-defined conditions, i.e. without possible interference of metabolism and of the dicarboxylate and tricarboxylate carriers, which also transport malate. The reaction mechanism of antiport, i.e. the sequence of binding and transport steps, can be investigated using the kinetic approach. From the mutual interaction of internal and external substrates with the carrier, it is possible to evaluate whether one (ping-pong mechanism) or two binding sites (sequential mechanism) have to be occupied before transport catalysis occurs. This type of analysis of a reconstituted mitochondrial carrier protein has successfully been carried out before in the case of the

Correspondence to F. Palmieri, Dipartimento Farmaco-Biologico, Università di Bari, Trav. 200 Re David, 4, I-70125 Bari, Italy

aspartate/glutamate carrier, leading to a sequential model of antiport [14]. Such two-substrate analysis was also applied to the oxoglutarate carrier using intact rat heart mitochondria [3]. Despite the complex kinetic behaviour of this carrier, as observed in the intact organelle (see above), the results obtained clearly indicated a sequential type of mechanism (rapid-equilibrium random).

When investigating the kinetic properties of the mitochondrial oxoglutarate carrier in a reconstituted system, two more basic aspects seemed also to be of interest. It was important to determine whether the data obtained using proteoliposomes were comparable to those measured in intact mitochondria, which would then imply that the conclusions drawn with respect to the mechanism of the carrier are also meaningful for the *in vivo* situation. Close similarity between mitochondria and the reconstituted system has recently been found in the case of the aspartate/glutamate carrier concerning its transport affinities as well as its kinetic mechanism [15]. Furthermore, in view of the growing structural family of mitochondrial carrier proteins [13, 16, 17], the question arises whether those carriers already studied kinetically would fall into the same category, also in terms of their intrinsic transport mechanism underlying antiport function, as has been suggested earlier [5].

MATERIALS AND METHODS

Materials

Hydroxyapatite (Bio-Gel HTP) and Dowex AG1-X8 were purchased from Bio-Rad, Celite 535 from Roth, Amberlite XAD-2 from Fluka, Sephadex G-75 from Pharmacia, 2-[5- ^{14}C]oxoglutarate and L-[U- ^{14}C]malate from Amersham, egg-yolk phospholipids (L- α -phosphatidylcholine from fresh turkey egg yolk), Pipes, bathophenanthroline and Triton X-114 from Sigma, pyridoxal 5'-phosphate from Boehringer Mannheim. All other reagents were of analytical grade.

Reconstitution of the oxoglutarate carrier

The oxoglutarate carrier was purified according to Bissaccia et al. [10]. Reconstitution of the carrier into liposomes was achieved by removing the detergent from mixed micelles containing Triton X-114, oxoglutarate carrier protein and egg-yolk phospholipids [18]. For this purpose the following mixture was subjected to hydrophobic chromatography on Amberlite beads; 320 μl purified carrier protein (10–20 μg) in 3% Triton X-114, 8.5 mg phospholipids in the form of sonicated liposomes prepared as described in [10], 2-oxoglutarate or L-malate at the indicated concentrations, 10 mM Pipes (pH 7) in a final volume of 700 μl . This mixture was passed 13 times through an Amberlite column (0.5 cm $\times$ 3.2 cm) equilibrated with a buffer containing 10 mM Pipes (pH 7) and the same concentration of the substrate present in the mixed micelles. The composition of the internal space of the proteoliposomes was thereby defined. All operations, except the hydrophobic chromatography, were performed at 4°C.

Transport measurement

In order to remove the external substrate, the proteoliposomes were passed through Sephadex G-75 columns (0.7 cm $\times$ 15 cm) equilibrated with 10 mM Pipes (pH 7) and an appropriate concentration of NaCl to balance the internal

osmolarity. The transport activity was determined by measuring the flux of ^{14}C -labelled substrates from outside to inside (forward exchange) or from inside to outside (backward exchange). For backward-exchange measurements, the internal substrate pool of the proteoliposomes was labelled by carrier-mediated equilibration with 0.5 μM [^{14}C]oxoglutarate or [^{14}C]malate added at high specific radioactivity; the internal substrate concentration was not changed (homologous exchange conditions). After 30 min, the radioactivity not taken up was removed by passing the proteoliposomes again through Sephadex G-75 (see above). The transport measurement was started, in the case of the backward exchange, by adding unlabelled substrate to the proteoliposomes or, in the case of the forward exchange, by adding ^{14}C -labelled substrate at the indicated concentrations. In either method, the exchange was stopped by addition of 50 mM pyridoxal 5'-phosphate and 10 mM bathophenanthroline, which were previously found to inhibit the oxoglutarate carrier [19, 20]. The incubation temperature was 25°C. Finally, each sample of proteoliposomes (100 μl) was applied to an anion-exchange column (Dowex AG1-X8; acetate form, 0.5 cm $\times$ 5 cm) in order to separate the external ^{14}C from the radioactivity remaining inside. The liposomes were eluted with 40 mM sodium acetate directly into the scintillation liquid.

Isotope-equilibration kinetics were followed by stopping the exchange after different time intervals. In the case of forward-exchange determinations, the transport rate was usually calculated from the radioactivity taken up by the proteoliposomes within 60 s. It was important for heterologous exchange experiments to stay as close as possible to initial conditions in order to avoid interference of the transported substrate in the opposite compartment. Faster (homologous exchange) kinetics, which were not linear for the first minute, were fitted by a computer program according to a first-order process equation allowing to calculate the initial transport rate. In the case of backward-exchange determinations, the decrease in radioactivity inside the liposomes until equilibrium was fitted according to a single exponential decay equation, from which the first-order rate constant, k , was derived. The backward-exchange rates were expressed as apparent velocities, v' , i.e. the product of k and the substrate concentration inside the liposomes, as described previously [21]. The values of v' are directly proportional to the actual transport rate. The proportionality factor depends on the portion of the total assay volume encapsulated by the actively exchanging proteoliposomes [21]. This factor can only be determined with low accuracy, but remains constant for each experiment. Therefore, the relative backward-exchange velocities were suitable for all kinetic investigations carried out in the present paper. It should be emphasized that the backward-exchange method was absolutely essential for measuring oxoglutarate carrier kinetics under particular conditions. The forward-exchange method is of limited applicability in experiments involving low internal concentration and, at the same time, external concentrations that are kinetically not limiting for the exchange measurement. In this case the isotope equilibrium that is reached, i.e. $\text{radioactivity}_{\text{in}}/\text{radioactivity}_{\text{ex}} = V_{\text{in}} \cdot [\text{S}]_{\text{in}}/V_{\text{ex}} \cdot [\text{S}]_{\text{ex}}$ (V , volume; $[\text{S}]$, substrate concentration; in, internal; ex, external), is drastically in favour of the external substrate pool ($V_{\text{ex}} \cdot [\text{S}]_{\text{ex}}$) due to the very low internal volume of the active proteoliposomes (V_{in}). Under these conditions, a more reliable signal can be monitored if the internal substrate pool is radioactively labelled allowing measurement of the export of radioactivity after addition of unlabelled substrate to the incubation medium.

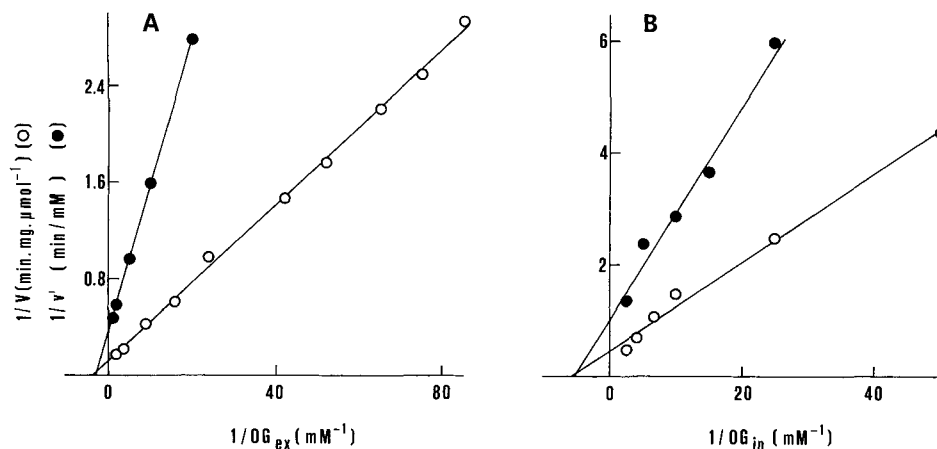


Fig. 1. K_m values of the reconstituted oxoglutarate carrier for oxoglutarate measured at the external (A) and internal (B) side of the liposomal membrane. Lineweaver-Burk plots were obtained from forward-exchange measurements of exchange velocity, v (○), or from backward-exchange measurements of rate constant, k , (●), under variation of external (A) or internal (B) oxoglutarate, as described in Materials and Methods. For definition of v' , see text. Substrate concentrations not indicated in the plots were as follows. (A) 20 mM (○) or 2 mM (●) internal oxoglutarate. (B) 0.15 mM (○) or 0.5 mM (●) external oxoglutarate. The K_m values from each plot were (A) 0.27 ± 0.06 mM by forward exchange and 0.34 ± 0.01 mM by backward exchange, and (B) 0.16 ± 0.04 mM by forward exchange and 0.18 ± 0.05 mM by backward exchange. OG_{ex} , external oxoglutarate; OG_{in} , internal oxoglutarate

Other methods

The protein concentration was determined by the Lowry method modified for the presence of Triton X-114 [22].

RESULTS

Validating the backward-exchange method

Antiport carriers catalyze a coupled counter exchange of internal and external substrate molecules across the membrane. Thus, for a complete kinetic analysis of antiport, the inward-facing binding site(s) has also to be described in terms of Michaelis constants of the transport reaction (K_m). This analysis, which has so far not been approached for the reconstituted oxoglutarate carrier, was made possible by using the backward-exchange technique (see Materials and Methods).

In Fig. 1 A, the internal substrate pool of proteoliposomes containing 2 mM oxoglutarate was radioactively labelled by carrier-mediated equilibration with a low external [^{14}C]-oxoglutarate concentration (0.5 μM) supplied at high specific radioactivity. During this step the internal substrate concentration remained unchanged (oxoglutarate/oxoglutarate homologous exchange). After removal of external radioactivity not taken up (see Materials and Methods), export of [^{14}C]-oxoglutarate was initiated by addition of various concentrations of unlabelled oxoglutarate (50–1000 μM). From the time course of the backward exchange, apparent first-order rate constants, k , were derived (see Materials and Methods) which were the basis for calculating apparent exchange velocities ($v' = k \cdot [S]_{\text{in}}$), as was elaborated previously for the reconstituted aspartate/glutamate carrier [21]. Fig. 1 validates the backward-exchange technique also for the reconstituted oxoglutarate carrier, as demonstrated by the external K_m for oxoglutarate determined both by the forward- and backward-exchange methods (Fig. 1 A). In either case, a straight line was obtained in Lineweaver-Burk plots over a wide concentration range, both lines showing the same intercept on the abscissa, which represents the reciprocal of the K_m . It should be pointed out that in Fig. 1 A the internal substrate concentration, although being different, was always clearly saturating (cf.

Table 1). The K_m obtained from this and several other experiments was 0.26 ± 0.12 mM, when derived from the backward-exchange measurements (Table 1), which agreed well with the result of the forward-exchange determinations (0.31 ± 0.08 mM, Table 1). It has to be pointed out that the different intercepts on the ordinate, as shown for forward- and backward-exchange measurements, do not represent different V_{max} values, since the backward-exchange activities are given as apparent velocities. From the forward-exchange data of Fig. 1 A, a V_{max} of $8.3 \mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$, was extrapolated.

Determination of internal transport affinities

The backward-exchange was therefore the appropriate method for determining the internal K_m for oxoglutarate. In the Lineweaver-Burk plot of Fig. 1 B, a linear dependence of the reciprocal of the exchange velocity on the reciprocal of the internal substrate concentration was observed even at concentrations as low as 40–400 μM , keeping external oxoglutarate constant at 0.5 mM, i.e. about twice its K_m . From 10 experiments of this type a K_m of 0.17 ± 0.06 mM (Table 1) was calculated for the internal binding site of the carrier. In order to confirm this value by forward-exchange measurements, the external oxoglutarate had to be decreased to a concentration below its K_m (0.15 mM) for methodological reasons as explained above. Nevertheless, a very similar K_m was obtained (0.20 ± 0.07 mM; Table 1).

For a complete kinetic description, we also had to determine the K_m values for malate, the second substrate of the oxoglutarate carrier. For this purpose, malate was substituted for oxoglutarate only in one compartment, keeping oxoglutarate as the constant counter substrate (heterologous exchange). Since the internal K_m for malate turned out to be clearly higher than the value obtained for oxoglutarate, we chose only the forward-exchange technique for further analysis, since this method gave reliable transport activity determinations at internal concentrations higher than 0.1 mM. The K_m values evaluated for malate were 0.71 ± 0.18 mM inside of the liposomes and 1.36 ± 0.14 mM outside (Table 1).

Table 1. K_m values of the mitochondrial oxoglutarate carrier: comparison of native and reconstituted protein. The values given are means $\pm$ SEM (n). n. d., not determined.

Substrate	Membrane side	Counter substrate	K_m		
			proteoliposomes		mitochondria
			forward exchange	backward exchange	
			mM		
Oxoglutarate	external	oxoglutarate malate	0.31 ± 0.08 (7) 0.28 ± 0.07 (3)	0.26 ± 0.12 (4) n. d.	$0.010-0.017^a$
	internal	oxoglutarate malate	0.20 ± 0.07 (3) 0.1	0.17 ± 0.06 (10) n. d.	1.0^b
Malate	external	oxoglutarate malate	1.36 ± 0.14 (6) 1.4	n. d. n. d.	0.21^c
	internal	oxoglutarate malate	0.71 ± 0.18 (6) 0.6	n. d. n. d.	1.3^a

^a Data from [9].

^b Data from [5].

^c Data from [6].

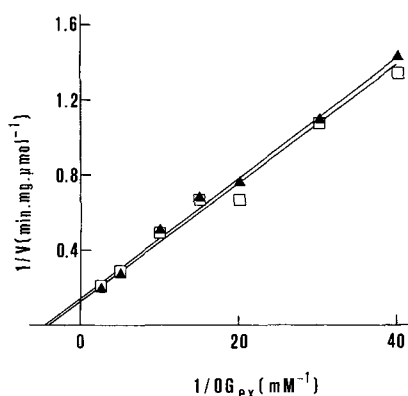


Fig. 2. Dependence of the kinetic constants of the oxoglutarate carrier on the substrate distribution. The K_m value for external oxoglutarate and the V_{max} were determined measuring uptake (forward exchange) of ^{14}C -labelled oxoglutarate into liposomes containing 20 mM oxoglutarate (□) or 20 mM malate (▲). The K_m values from each plot were 0.25 ± 0.06 mM (□) and 0.23 ± 0.04 mM (▲); the V_{max} values 7.8 ± 1.8 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$ (□) and 7.0 ± 1.2 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$ (▲). OG_{ex} , external oxoglutarate

Independence of antiport kinetics from substrate distribution

When investigating the kinetic mechanism of antiport, it is of basic interest to know whether the antiport activity depends on the substrate distribution on the two sides of the membrane, i.e. whether the same kinetic parameters are observed for one substrate on one side, independent of the nature of the counter substrate. To clarify this question, V_{max} values of oxoglutarate uptake were determined, not only in oxoglutarate/oxoglutarate homologous exchange, but also in heterologous exchange experiments (external oxoglutarate/internal malate). Either of the two sets of data shown in Fig. 2 fitted almost perfectly the same straight line, thus showing that not only the V_{max} but also the K_m of external oxoglutarate was not significantly changed, whether or not 20 mM oxo-

glutarate (V_{max} 7.8 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$; K_m 0.25 mM) or 20 mM malate (V_{max} 7.0 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$; K_m 0.23 mM) were present inside the liposomes. The same result was obtained in another experiment measuring the uptake of malate into liposomes containing 20 mM malate (V_{max} 11.0 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$; K_m 1.4 mM) or 20 mM oxoglutarate (V_{max} 9.2 $\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$; K_m 1.4 mM; experiments not shown). The observed independence of the reconstituted antiport activity from the nature of the respective substrate at the opposite membrane side was corroborated when determining the internal half-saturation constants in homologous or heterologous exchange experiments (see Table 1).

Orientation of the reconstituted carrier protein

In previous studies on the kinetics of the oxoglutarate translocator carried out in intact mitochondria [2, 3, 6, 8, 9], the K_m values for external substrates were reported to be significantly lower (cf. Table 1) than those shown above for either side of the liposomal membrane. This discrepancy becomes most obvious with respect to the K_m as found in mitochondria for external oxoglutarate [8], where, in a non-Michaelian saturation curve, numerical half-saturation was observed at $10 - 17$ μM , differing by at least one order of magnitude from our K_m values (internal; $0.17 - 0.20$ mM; external; $0.26 - 0.31$ mM). Concerning the outside face of the reconstituted oxoglutarate carrier, no additional K_m value for oxoglutarate could be detected down to concentrations as low as 12 μM (Fig. 1A). Thus, the alternative possibility had to be tested that a high-affinity binding site was overlooked at inside face of the reconstituted carrier, even though in Fig. 1B internal oxoglutarate concentrations were as low as 20 μM . Under the conditions of this experiment (Fig. 1B), i.e. the external concentration exceeding the internal by a factor of 10 ($\text{radioactivity}_{ex}/\text{radioactivity}_{in} = V_{ex} \cdot [S]_{ex}/V_{in} \cdot [S]_{in} \geq 5000$), isotope equilibration was extremely fast in both forward- and backward-exchange measurements. Therefore, it was necessary to decrease also the external substrate to a very low concentration (5 μM , Fig. 3A), which then allowed investigation of a wide range of internal substrate concentrations, as

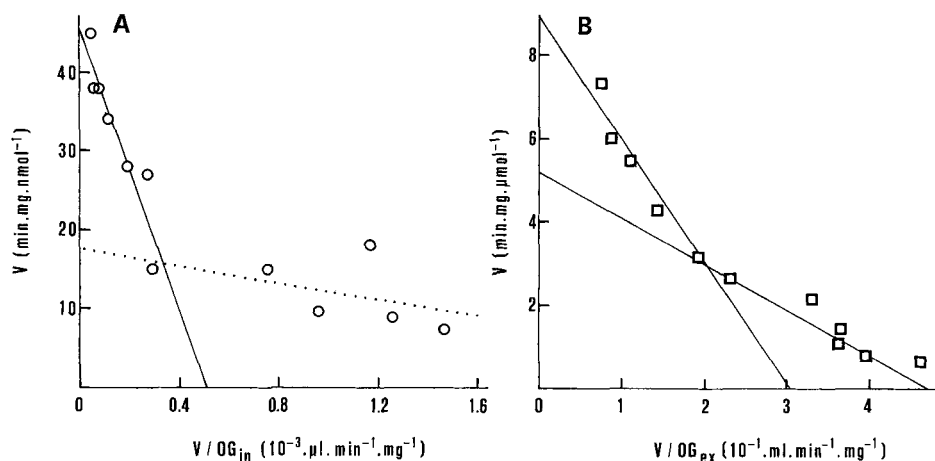


Fig. 3. Reshuffling of internal and external binding sites of the reconstituted oxoglutarate carrier after sonication of proteoliposomes. (A) Apparent K_m values (see text) for oxoglutarate were measured at the internal membrane side by the forward-exchange method without sonication of the liposomes. The K_m values were calculated from the slope of the Eadie-Hofstee plots. The relevance of the dotted line is described in the text. (B) After freezing and thawing the proteoliposomes followed by a 3-min pulse sonication, the K_m values for oxoglutarate were measured at the external membrane side by the forward-exchange method. Two K_m values were found, one (0.1 mM) corresponding to the internal K_m (A) and the other (0.28 mM) corresponding to the external K_m observed in Fig. 1A. The oxoglutarate concentrations were as follows: (A) 5 μ M external oxoglutarate, 5–1000 μ M internal oxoglutarate. (B) 20 mM internal oxoglutarate, 5–1000 μ M external oxoglutarate. The K_m values derived from each plot were (A) 100 ± 15 μ M and 5.4 ± 4.3 μ M for internal oxoglutarate and (B) 0.28 ± 0.03 mM and 0.10 ± 0.03 mM for external oxoglutarate. The transport was assayed by the forward-exchange procedure, as described in Materials and Methods, for the case of non-linear (A) or linear (B) uptake in the first minute. OG_{in} , internal oxoglutarate; OG_{ex} , external oxoglutarate

for oxoglutarate (5–1000 μ M, Fig. 3A) and malate (15–3000 μ M, not shown). In these experiments in fact, an additional kinetic component representing a low apparent K_m was revealed for internal oxoglutarate (about 5 μ M) and, surprisingly, also for malate (about 25 μ M). The two K_m values, 5 μ M and 0.1 mM, for internal oxoglutarate, were calculated from the different slopes of the two lines in the Eadie-Hofstee plot shown in Fig. 3A. This type of plot was chosen for a better resolution of the results. As estimated from the ordinate intercepts, the high-affinity component would contribute about 40% of the overall maximum activity.

These low internal K_m values, however, could be due to artefacts resulting from the lack of pseudo-first-order conditions or, in other words, from the lack of steady-state conditions. This can be rationalized assuming one or more carrier molecules/active proteoliposome containing very few, i.e. less than 10, substrate molecules at 20 μ M, as can be calculated theoretically. To confirm or deny, nevertheless, the existence of high-affinity binding sites at the internal surface of the carrier, the proteoliposomes, after formation, were sonicated in order to scramble the orientation of the carrier molecules. After this procedure, we could expect to find the low K_m also outside the liposomes, where it could be measured without experimental restrictions. Fig. 3B shows the transport rate as a function of external oxoglutarate (5–1000 μ M) after sonication of proteoliposomes (20 mM oxoglutarate inside). As in Fig. 3A, a non-linear dependence became evident in an Eadie-Hofstee plot. However, the two different K_m values observed, 0.28 mM and 0.1 mM, corresponded to the external K_m found previously (Fig. 1A) and to the high internal K_m (cf. Fig. 3A), respectively. No additional K_m in the micromolar range could be found that would contribute significantly, i.e. more than 15%, to the overall maximum velocity. Instead, the V_{max} was divided into two portions of almost the same size by the two kinetic components. Obviously the orientation of the membrane-inserted carrier molecules was randomized by the sonication treatment.

In order to obtain a second criterion clarifying the question of asymmetric carrier orientation, we furthermore investigated whether the internal K_m value for malate (0.7 mM) could also be found on the outside of the liposomes and whether the external K_m for malate (1.4 mM) could be found on the inside. The factor of two difference between these two values promised results of better reliability than the difference between the respective K_m values for oxoglutarate. Fig. 4 shows that again a K_m of about 1.4 mM for external malate was found, although the substrate was now varied in the concentration range of the internal K_m , 0.03–1.2 mM. Consequently, no additional K_m below 1 mM was present at the outside face of the proteoliposomes. Likewise no additional K_m above 1 mM was detected when investigating the internal side of the proteoliposomes, covering a concentration range of 0.25–4 mM (K_m 0.65 mM, Fig. 4). Although not absolutely convincing, the relatively low statistical error of the two K_m values (see Fig. 4), however, indicates a true difference between the internal and external binding sites. As already mentioned above, this result was independent whether oxoglutarate or malate was used as internal counter substrate (not shown).

Kinetic mechanism of antiport

The asymmetric orientation of the membrane-embedded carrier on the one hand and the independence of the antiport reaction from the type of counter substrate on the other hand, provided well-defined conditions to study the transport mechanism by means of two-reactant initial-velocity studies. The counter exchange of substrates catalyzed by an antiport carrier can be described in terms of a two-substrate reaction involving different steps of binding, transport and release of substrates. The sequence of these steps follows one of two basically different types of two-substrate mechanism, the ping-pong or the sequential type [23].

In order to discriminate between these two possibilities for the reconstituted oxoglutarate carrier, the antiport activity

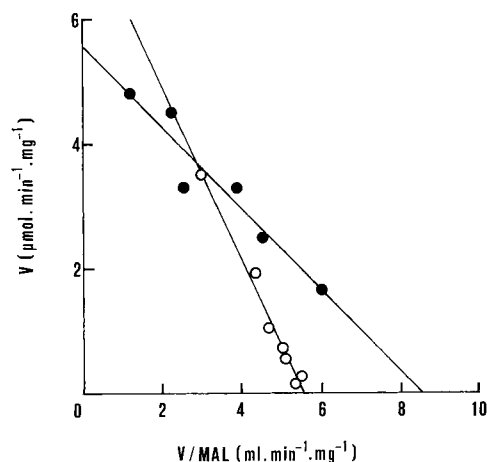


Fig. 4. Functional analysis of the carrier orientation. In order to elucidate the sidedness of the binding sites characterized by different K_m values, the K_m for external malate (○) was determined varying the external malate concentration over the range 0.03–1.2 mM, i.e. in the range of the observed internal K_m for malate (0.7 mM). Only one external K_m of 1.38 ± 0.08 mM could be found. The difference to the internal side of the carrier is demonstrated by the determination of the internal K_m for malate (●), obtained by varying the internal malate concentration over the range 0.25–4 mM, i.e. in the range of the observed external K_m for malate (1.4 mM). Only one internal K_m of 0.65 ± 0.10 mM could be found. All data were obtained by the forward-exchange mode; (○) 20 mM internal oxoglutarate; (●) 0.5 mM external oxoglutarate. MAL, malate

had to be analyzed while varying the substrate concentrations in both the internal and the external compartments within the same experiment. Fig. 5 shows the results of such a two-reactant initial-velocity study (two-substrate analysis) of the oxoglutarate/oxoglutarate homologous exchange reaction, carried out applying the backward-exchange method. When the kinetic data were analyzed in Lineweaver-Burk plots, showing the dependence of the transport rate on internal oxoglutarate (40–400 μM) at three different external concentrations (40, 120 and 500 μM), a pattern of straight lines was obtained intersecting at a common point on the abscissa (Fig. 5A). If the same data were plotted as a function of the external oxoglutarate concentration (Fig. 5B), a similar pattern was obtained. This intersecting pattern, opposed by the parallel pattern of ping-pong reactions, is indicative of a sequential type of mechanism [23], which is observed whenever the two substrate molecules bind before catalysis occurs. Thus, the oxoglutarate carrier had to form a ternary complex with the two counter substrates prior to translocation. The intersection point on the abscissa shows that there was little or no influence of the first, i.e. internal (or external), substrate on the apparent K_m for the second, i.e. external (or internal), substrate. Only the transport rate was stimulated with increasing concentration of the counter substrate, as becomes apparent from the different apparent $V_{\max}$ values (reciprocal ordinate intercepts) and, consequently, also from the different slopes representing apparent K_m /apparent $V_{\max}$. This mutual dependence could be quantitatively analyzed in replots of the slope and the ordinate intercepts of the primary curves vs. the reciprocal concentration of the so-called non-varied substrate. This was carried out in Fig. 5C and D referring to the data of Fig. 5A and B. From the linear relations obtained, K_m values could be derived that were extrapolated for infinite concentration of the respective second substrate (cf. [14]).

These concentration-independent K_m values, 0.19 mM for external and 0.15 mM for internal oxoglutarate, agreed well with the values determined above in backward-exchange measurements (Fig. 1) at finite substrate concentration (0.26 mM and 0.17 mM, respectively; see Table 1). Furthermore, the K_i values, representing the dissociation constants of the binary carrier-oxoglutarate complexes in a sequential mechanism [23, 24], could be extracted from the secondary plots. The values obtained independently from Fig. 5C and D, 0.15 mM for external and 0.12 mM for internal oxoglutarate, were very similar to the respective K_m values. This leads to a constant factor between the K_m and K_i values, i.e. 0.79 and 0.80 for external and internal oxoglutarate, respectively, which is in agreement with a rapid-equilibrium random mechanism.

It was important to determine whether this particular mechanism also holds true for the malate/malate-exchange reaction. Due to the higher internal K_m for malate than oxoglutarate, this question could be approached by using the forward-exchange technique (see above), which is less complex than the backward exchange. In a first step, however, the forward exchange was applied to reexamine the oxoglutarate/oxoglutarate homologous exchange in order to substantiate the results obtained above by this independent method (Fig. 6A). Although in this two-substrate analysis we were not able to decrease the internal oxoglutarate to levels clearly below the internal K_m , as was possible in the backward-exchange experiments (Fig. 5), again a kinetic pattern of the sequential type was obtained showing an intersection point on the abscissa. Applying secondary plots (not shown) a maximum velocity could be extrapolated for infinite concentration of oxoglutarate in both compartments ($8.3 \mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$). The concentration-independent K_m values determined were less reliable than those obtained in Fig. 5C and D, since external oxoglutarate could only be varied below its K_m . The internal value was 0.11 mM, hence confirming the backward-exchange results. Finally, the malate/malate-exchange reaction was investigated as shown in Fig. 6B. Also in this case an intersection point very close to the abscissa was observed demonstrating that the reaction mechanism was not altered when a different substrate was translocated ($V_{\max}$ $8.5 \mu\text{mol} \cdot \text{min}^{-1} \cdot \text{mg protein}^{-1}$; K_m for internal malate 0.63 mM, for external malate 1.2 mM). The K_i values, like those of the oxoglutarate/oxoglutarate exchange, fulfilled the correlation K_i/K_m (internal) = K_i/K_m (external), as expected for a random mechanism [23, 24].

DISCUSSION

Kinetic description of the reconstituted oxoglutarate carrier and comparison to mitochondria

The liposomal system, in principle, allows to free variation of substrate concentration not only at the external but also at the internal side of the membrane, an obvious and important advantage in comparison to the much more difficult experiments using intact organelles. Furthermore it was possible to use both the forward- and the backward-exchange methods, the compatibility of which has been demonstrated in Results (see Table 1). For particular experimental conditions (see Materials and Methods) one of the two methods led to more reliable kinetic data; in combination, forward- and backward-exchange measurements led to results reinforcing each other. We thereby succeeded in obtaining all kinetic constants necessary to describe the two different sides of the reconstituted oxoglutarate carrier (Table 1) showing medium-affinity bind-

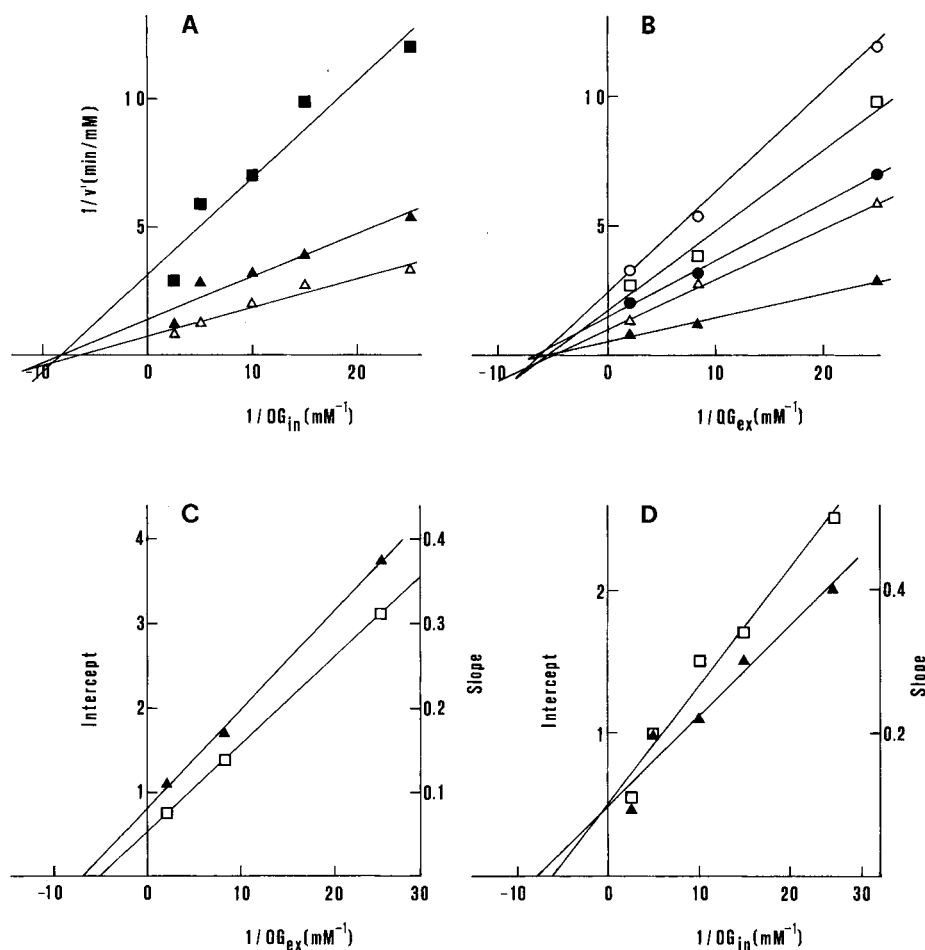


Fig. 5. Two-substrate analysis by the backward-exchange method of the oxoglutarate/oxoglutarate homologous exchange reaction mediated by the reconstituted oxoglutarate carrier. (A, B) Lineweaver-Burk plots showing the dependence of exchange activity on internal (A) and external (B) oxoglutarate. The concentrations of the counter substrates were as follows. (A) 0.04, (■), 0.12, (▲) and 0.5 (△) mM external oxoglutarate; (B) 0.04 (○), 0.067 (□), 0.1 (●), 0.2 (△) and 0.4 (▲) mM internal oxoglutarate. Apparent exchange rates v' were calculated as described in Materials and Methods for the backward-exchange procedure. (C, D) Slope (▲) and intercept (□) replots of the primary plots A and B, respectively. The concentration-independent K_m values and the K_i values (see text) were derived from the intercepts and slopes of the secondary plots [22]: K_m (external) 0.19 ± 0.001 mM (C) and 0.19 ± 0.1 mM (D); K_m (internal) 0.15 ± 0.02 mM (C) and 0.16 ± 0.04 mM (D); K_i (external) 0.15 ± 0.04 mM (D); K_i (internal) 0.12 ± 0.01 mM (C). OG_{in} , internal oxoglutarate; OG_{ex} , external oxoglutarate

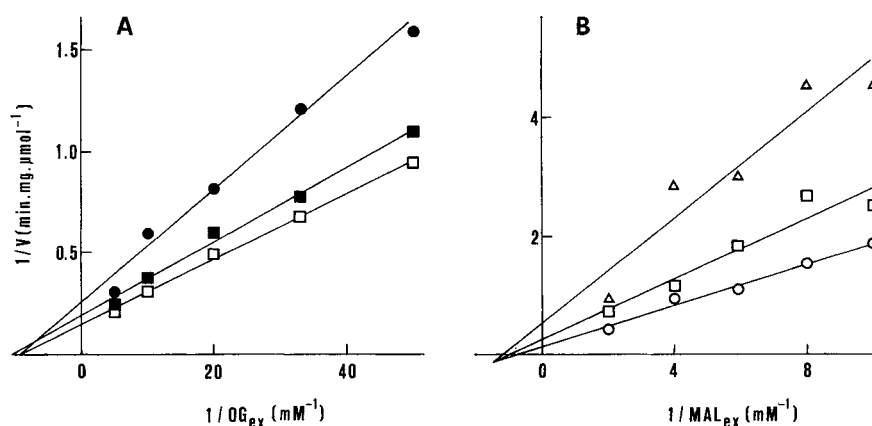


Fig. 6. Two-substrate analysis by the forward-exchange method of the oxoglutarate/oxoglutarate and malate/malate homologous exchange reactions mediated by the reconstituted oxoglutarate carrier. Lineweaver-Burk plots showing the dependence of exchange activity on external oxoglutarate (A) and external malate (B). The concentrations of the counter substrates were as follows (A) 0.1 (●), 0.2 (■), 0.4 (□) mM internal oxoglutarate; (B) 0.2 (△), 0.6 (□), and 5 (○) mM internal malate. Exchange rates were calculated as described in Materials and Methods for the forward-exchange procedure. OG_{ex} , external oxoglutarate; MAL_{ex} , external malate

ing at the inside (K_m 0.2 mM and 0.7 mM for oxoglutarate and malate, respectively) and medium-to-low-affinity binding at the outside of the liposomes (K_m 0.3 mM and 1.4 mM). When comparing the kinetic data obtained here with those from detailed studies in intact mitochondria [2–9, 25], it becomes obvious that the situation in the proteoliposomes is definitely less complicated. We did not observe cooperativity phenomena and, due to the absence of other metabolites, no possible allosteric influences had to be considered (see above). The reciprocal plots were linear both for malate and for oxoglutarate on either membrane side, whereas in rat heart mitochondria a complex pattern was observed for external oxoglutarate, which was interpreted as multiple binding sites [4–6, 9]. Nevertheless, the basic kinetic properties are in reasonably good agreement, disregarding, however, the affinity for cytosolic oxoglutarate (see below). It was shown before that the substrate specificity and the inhibitor sensitivity of the reconstituted and the native translocator are very similar [10–12]; furthermore, the kinetic mechanism proved to be identical (see below). Presumably, the drastically reduced kinetic complexity, compared to the observations in mitochondria, reflects the reduced complexity of the liposomal system. For reconstitution, we used a more or less uncharged membrane, inserting a single protein species in a few copies/liposome. Thus, specific protein-protein and protein-lipid interactions, which could be responsible for the surprisingly complex kinetic behaviour under physiological conditions, do not play any role in the liposomal system.

When studying a reconstituted carrier, it is important to know whether the protein is oriented unidirectionally within the liposomal membrane, unless the two possible orientations can be differentiated by side-specific inhibitors. Unfortunately, such inhibitors are only available for the adenine nucleotide carrier [26]. Eosine maleimide, which was shown to inhibit the oxoglutarate carrier only at the cytosolic side of mitochondria [27], led to complete inhibition of the exchange in the reconstituted system. However, the liposomal membrane proved to be permeable to this reagent (unpublished results). Unidirectionality of membrane insertion has previously been elucidated in the case of the reconstituted aspartate/glutamate carrier by determining the sidedness of external and internal binding sites characterized by different affinity constants [21]. In the present study, we used the same approach for the reconstituted oxoglutarate carrier.

The results of Table 1 show that the apparent K_m values on the inside are different from those on the outside. These differences are obviously not very significant in the case of oxoglutarate. For internal and external malate, however, the statistical error is low enough for reliable discrimination. As well as this result, two independent lines of evidence are in favour of distinct K_m values on the two opposite sides. When applying two independent methods of analysis, the same ratio of external and internal K_m for oxoglutarate was obtained in forward-exchange (1.55) and backward-exchange (1.53) determinations. Furthermore, when scrambling the orientation of the carrier by sonication of the proteoliposomes, in fact both expected K_m values were found at the external side in approximately 1:1 stoichiometry (Fig. 3B). The sidedness of the two different affinity constants, without sonication, is demonstrated in Figs 1 and 2 for oxoglutarate and in Fig. 4 for malate.

These results indicate that the oxoglutarate carrier is predominantly oriented unidirectionally in the liposomal membrane. Unfortunately we could not determine unequivocally, whether this orientation is right-side out or inside out in com-

parison to intact mitochondria. Although the K_m values measured in the reconstituted system, with one exception, were not drastically different from those found in mitochondria (Table 1), no conclusive correlation of one mitochondrial side with one liposomal side became apparent due to the small differences between the internal and external affinities found in either system. The only exception found by Sluse et al. [9] was the very low half-saturating concentration of external oxoglutarate (10–17 μ M), which in rat heart mitochondria corresponded to an intermediate plateau of a complex saturation curve (see above). It should be noted, however, that there exist conflicting results concerning the possibility to achieve complete saturation of external binding sites by oxoglutarate in intact mitochondria [4, 6, 9, 28]. In proteoliposomes we determined a single K_m at each side, namely 0.3 mM for external and 0.2 mM for internal oxoglutarate.

As already discussed in Results, the observation of the low internal K_m was very likely to be an experimental artefact, mainly due to the extremely low amount of internal substrate under the conditions of Fig. 3A, leading to non-steady-state kinetics. There are further arguments against the validity of the low apparent K_m . Although in mitochondria a low (external) K_m was reported only for oxoglutarate and not for malate, we observed the same putative artefact, i.e. a very low K_m (20–25 μ M, not shown), also when using internal malate in comparable low concentrations. Furthermore, after scrambling the membrane topology of the proteoliposomes by sonication, only the normal inside K_m (0.1 mM, Fig. 3) appeared at the outer surface of the liposomes.

There are obvious explanations for the differences in the apparent affinities between proteoliposomes and mitochondria. The source of the oxoglutarate carrier was different, rat heart in the case of intact mitochondria and bovine heart in the case of the reconstituted system. Furthermore, the kinetic analysis of Sluse et al. [9] was carried out at 2°C, whereas we measured exchange in proteoliposomes at room temperature in order to have sufficient activity of a drastically reduced concentration of carrier protein in comparison to mitochondria. Palmieri et al. [2] already observed in rat liver mitochondria that the external K_m for oxoglutarate significantly increases from 20 μ M to 46 μ M when shifting the temperature from 0°C to 9°C.

In summary, our results concerning the carrier orientation are indicative, although not absolutely convincing, of an unidirectional insertion of the oxoglutarate carrier into liposomal membrane. However, it could not be decided whether the orientation of the reconstituted protein is right-side out or inside out compared to mitochondria. On the basis of the most relevant K_m found in rat heart mitochondria, i.e. the K_m for external malate which was the only Michaelian substrate at the cytosolic side [6], one could speculate on an inside-out orientation.

The sequential antiport mechanism, a common characteristic of mitochondrial carriers

Fortunately, with respect to the kinetic mechanism, the situation is not at all ambiguous. As found previously in rat heart mitochondria [3], two-reactant initial-velocity studies clearly revealed a sequential type of mechanism, as documented for the reconstituted oxoglutarate carrier in Figs 5 and 6. The intersecting lines in the reciprocal plots and the definite slope obtained in secondary plots (slope replot, Fig. 5C and D) can only be interpreted in terms of this kinetic mechanism [23, 24] which implies the occurrence of a ternary complex

(carrier molecule with two substrate molecules bound) during the catalytic cycle of antiport. The common intersection point on the abscissa suggests a random mechanism of substrate binding. This means that there is no obligatory order in the binding of the two counter substrates. Moreover, the first substrate, bound at one side, does not exert significant influence on the apparent affinity for the second substrate which is bound from the other side. Further observations are in line with this conclusion. First, the concentration-independent K_m values, obtained by extrapolation from secondary plots, agree well with the apparent K_m values at finite substrate concentration. Second, the K_i values, representing substrate interaction with the unloaded carrier, more or less coincide with the corresponding K_m values. Thus, the affinity is almost identical whether binding occurs to the free or to the single-substrate-occupied carrier. These results are in agreement with a special case of a sequential mechanism, i.e. a rapid-equilibrium random mechanism with fast and independent binding of the two substrate molecules. The values for K_i in a ping-pong mechanism would be zero, since no ternary complex is involved. The independent interaction of the two substrates with the translocator protein was observed not only for oxoglutarate (Fig. 5) but also for malate homologous exchange (Fig. 6B), as well as heterologous exchange involving different counter substrates (not shown). An absolutely identical kinetic behaviour of the oxoglutarate carrier was observed in all cases.

When summarizing the following observations, (a) the necessity of a ternary complex, (b) the independence of internal and external binding sites, (c) no influence of the different substrates on the transport rate, and (d) the previous observation of a competitive behaviour of oxoglutarate and malate when present at the same side [12], a functional model for the oxoglutarate carrier becomes evident. This model implies a ternary complex with one substrate molecule bound at either side of the carrier as the catalytic, i.e. transport-competent, intermediate, exchanging the two substrates in a concerted step between the two compartments. The symmetrical model closely resembles that described in detail for the reconstituted aspartate/glutamate carrier [14]. However, in the case of the oxoglutarate carrier no differently charged substrate-carrier complexes have to be considered arising from H^+ cotransport.

As discussed above, a sequential type of kinetic mechanism has already been shown before for the oxoglutarate carrier in rat heart mitochondria [3] and for the aspartate/glutamate carrier both in the reconstituted system [14] and in mitochondria [15, 29]. Since there is increasing evidence from studies in intact mitochondria that also the ADP/ATP carrier functions according to this type of mechanism [5], a general concept seems to emerge, where the close relationship of the members of the mitochondrial carrier family is not only expressed in the similarity of the primary structure and conformation, but also in a common type of carrier mechanism.

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REFERENCES

1. Quagliariello, E. & Palmieri, F. (1972) in *Biochemistry and biophysics of mitochondrial membranes* (Azzone, G. F., Carafoli, E., Lehninger, A. L., Quagliariello, E. & Siliprandi, N., eds) pp. 659–680, Academic Press, New York.
2. Palmieri, F., Quagliariello, E. & Klingenberg, M. (1972) *Eur. J. Biochem.* **29**, 408–416.
3. Sluse, F. E., Ranson, M. & Liébecq, C. (1972) *Eur. J. Biochem.* **25**, 207–217.
4. Sluse-Goffart, C. M. & Sluse, F. E. (1986) in *Dynamics of biochemical systems* (Damjanovic, S., Keleti, T. & Tron, L., eds), pp. 521–535, Elsevier, Amsterdam.
5. Sluse, F. E., Sluse-Goffart, C. M. & Duyckaerts, C. (1989) in *Anion carriers of mitochondrial membranes* (Azzi, A., Nalecz, K. A., Nalecz, M. J. & Wojtczak, L., eds) pp. 183–195, Springer Verlag, Berlin, Heidelberg, New York.
6. Sluse-Goffart, C. M., Sluse, F. E., Duyckaerts, C., Richard, M., Hengesch, P. & Liébecq, C. (1983) *Eur. J. Biochem.* **134**, 397–406.
7. Sluse, F. E., Sluse-Goffart, C. M., Duyckaerts, C. & Liébecq, C. (1975) *Eur. J. Biochem.* **56**, 1–14.
8. Sluse, F. E., Duyckaerts, C., Sluse-Goffart, C. M., Fux, J. P. & Liébecq, C. (1980) *FEBS Lett.* **120**, 94–98.
9. Sluse, F. E., Duyckaerts, C., Liébecq, C. & Sluse-Goffart, C. M. (1979) *Eur. J. Biochem.* **100**, 3–17.
10. Bisaccia, F., Indiveri, C. & Palmieri, F. (1985) *Biochim. Biophys. Acta* **810**, 362–369.
11. Bisaccia, F., Indiveri, C. & Palmieri, F. (1988) *Biochim. Biophys. Acta* **933**, 229–240.
12. Indiveri, C., Palmieri, F., Bisaccia, F. & Krämer, R. (1987) *Biochim. Biophys. Acta* **890**, 310–318.
13. Runswick, M. J., Walker, J. E., Bisaccia, F., Iacobazzi, V. & Palmieri, F. (1990) *Biochemistry* **29**, 11033–11040.
14. Dierks, T., Riemer, E. & Krämer, R. (1988) *Biochim. Biophys. Acta* **943**, 231–244.
15. Evens, A., Duyckaerts, C., Sluse-Goffart, C. M., Dierks, T., Krämer, R. & Sluse, F. E. (1990) *Eur. Bioenerg. Conf. Rep.* **6**, S.III–35.
16. Runswick, M. J., Powell, S. J., Nyren, P. & Walker, J. E. (1987) *EMBO J.* **6**, 1367–1373.
17. Aquila, H., Link, T. A. & Klingenberg, M. (1987) *FEBS Lett.* **212**, 1–9.
18. Krämer, R. & Heberger, C. (1986) *Biochim. Biophys. Acta* **868**, 289–296.
19. Indiveri, C., Krämer, R. & Palmieri, F. (1987) *J. Biol. Chem.* **262**, 15979–15984.
20. Passarella, S., Palmieri, F. & Quagliariello, E. (1973) *FEBS Lett.* **38**, 91–95.
21. Dierks, T. & Krämer, R. (1988) *Biochim. Biophys. Acta* **937**, 112–126.
22. Dulley, J. R. & Grieve, P. A. (1975) *Anal. Biochem.* **64**, 136–141.
23. Cleland, W. W. (1970) in *The enzymes* (Boyer, P.D., ed.) vol. 2, pp. 1–65, Academic Press, New York.
24. Segel, I. H. (1975) *Enzyme kinetics*, pp. 274–320, Wiley, New York.
25. Sluse, F. E., Goffart, G. & Liébecq, C. (1973) *Eur. J. Biochem.* **32**, 283–291.
26. Krämer, R. & Klingenberg, M. (1979) *Biochemistry* **18**, 4209–4215.
27. Zara, V. & Palmieri, F. (1988) *FEBS Lett.* **236**, 493–496.
28. Duyckaerts, C., Sluse-Goffart, C. M., Sluse, F. E., Gosselin-Rey, C. & Liébecq, C. (1984) *Eur. J. Biochem.* **142**, 203–208.
29. Murphy, E., Coll, K. E., Viale, R. O., Tischler, M. E. & Williamson, J. R. (1979) *J. Biol. Chem.* **254**, 8369–8376.